Prediction accuracy — heart_attack_youngsters_india.csv

1. Test-set performance

Model	CV ROC-AUC (mean ± SD)	Test ROC-AUC (95% CI)	Acc.	Sens.	Spec.	F1	MCC	Brier
Soft voting	**0.997 ± 0.002**	**1.000 (0.999–1.000)**	**99.0%**	**0.991**	**0.988**	**0.991**	**0.979**	**0.008**
Stacking	**0.997 ± 0.002**	**1.000 (0.999–1.000)**	**99.0%**	**0.991**	**0.988**	**0.991**	**0.979**	**0.008**
LightGBM	0.996 ± 0.002	1.000 (0.999–1.000)	99.0%	0.991	0.988	0.991	0.979	0.006
EBM	0.996 ± 0.002	1.000 (0.998–1.000)	99.5%	1.000	0.988	0.996	0.990	0.005
XGBoost	0.994 ± 0.005	0.999 (0.998–1.000)	98.5%	0.991	0.976	0.987	0.969	0.009
CatBoost	0.996 ± 0.002	0.999 (0.998–1.000)	99.0%	0.991	0.988	0.991	0.979	0.007
Random forest	0.996 ± 0.001	0.999 (0.997–1.000)	99.0%	0.983	1.000	0.991	0.980	0.019
Logistic regression	0.991 ± 0.004	0.999 (0.996–1.000)	98.5%	0.983	0.988	0.987	0.969	0.013

2. Cross-validation on the development set

Model	ROC-AUC	PR-AUC	Accuracy	Sensitivity	Specificity	MCC	Brier
Soft voting	**0.997**	**0.998**	**96.8%**	**0.970**	**0.964**	**0.933**	**0.020**
Stacking	**0.997**	**0.998**	**96.8%**	**0.970**	**0.964**	**0.933**	**0.021**
LightGBM	0.996	0.997	96.5%	0.959	0.973	0.929	0.020
EBM	0.996	0.997	97.5%	0.983	0.964	0.949	0.019
XGBoost	0.994	0.995	95.7%	0.959	0.955	0.914	0.029
CatBoost	0.996	0.997	96.1%	0.959	0.964	0.921	0.022
Random forest	0.996	0.997	96.0%	0.961	0.959	0.919	0.027
Logistic regression	0.991	0.993	96.4%	0.968	0.958	0.926	0.029

3. Calibration and operating points

Model	Youden threshold	Cal. intercept	Cal. slope	Sens ≥ 0.90 threshold	Sensitivity there	Specificity there
Soft voting	0.523	-0.13	1.84	0.826	0.983	1.000
Stacking	0.536	0.00	1.32	0.955	0.983	1.000
LightGBM	0.499	-0.17	0.88	0.986	0.966	1.000
EBM	0.373	-0.44	1.03	0.976	0.948	1.000
XGBoost	0.295	-0.03	1.12	0.966	0.931	1.000
CatBoost	0.614	-0.45	1.17	0.952	0.957	0.988
Random forest	0.637	-0.09	3.29	0.763	0.957	1.000
Logistic regression	0.460	-0.04	1.44	0.861	0.940	0.988

4. Figures

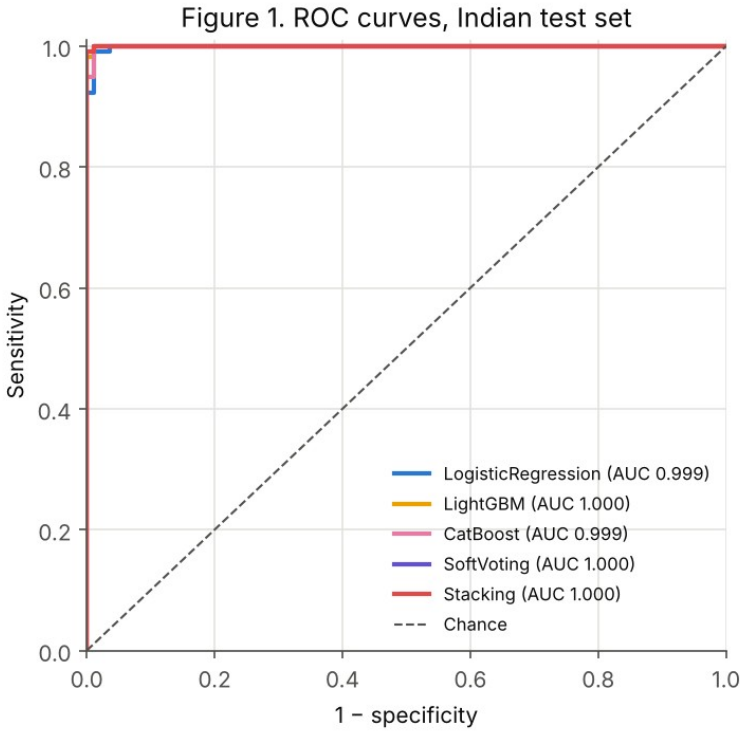


Figure 1. ROC curves on the test set

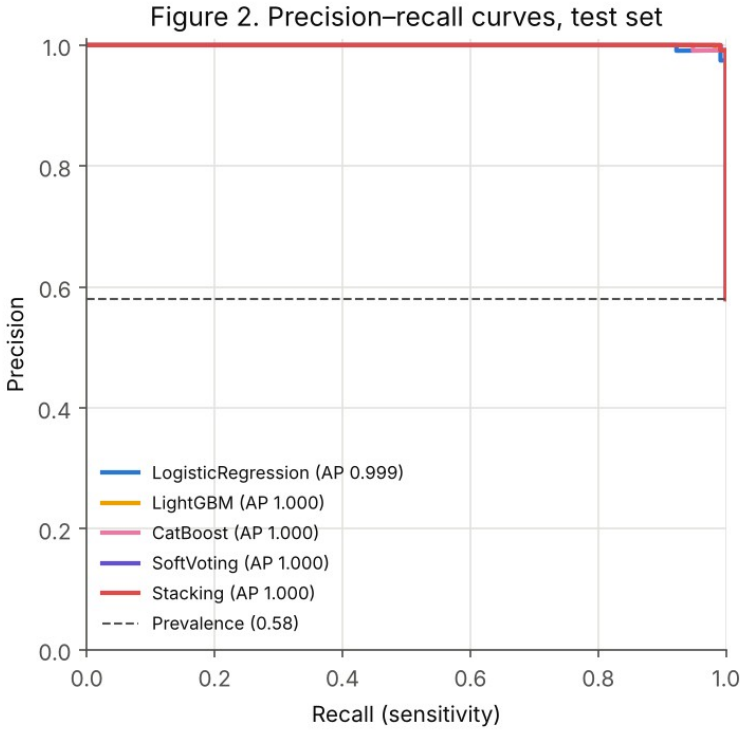


Figure 2. Precision–recall curves on the test set

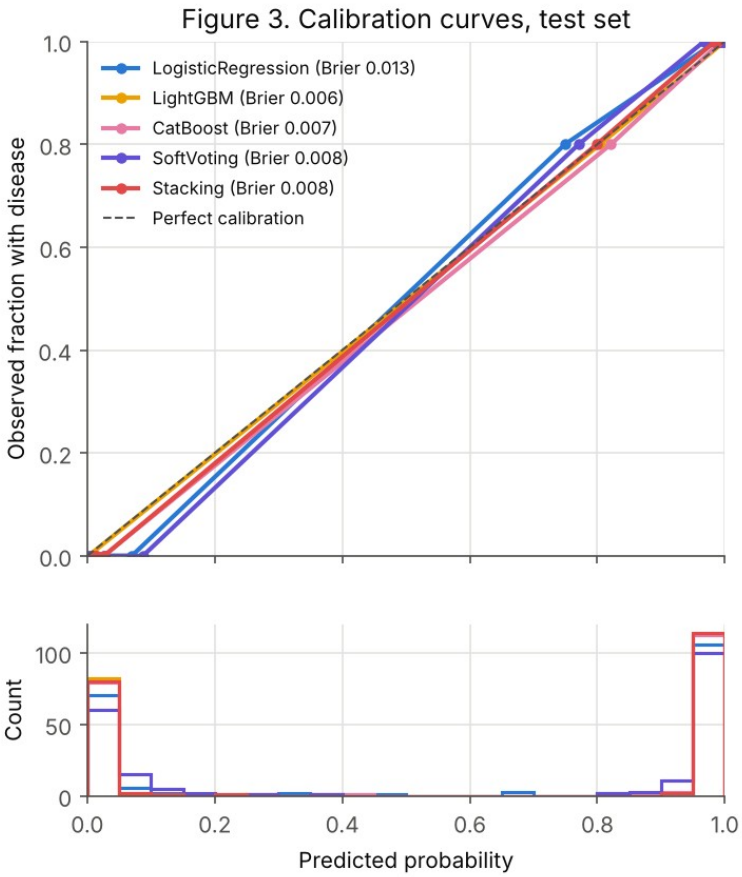


Figure 3. Calibration curves and distribution of predicted probabilities on the test set

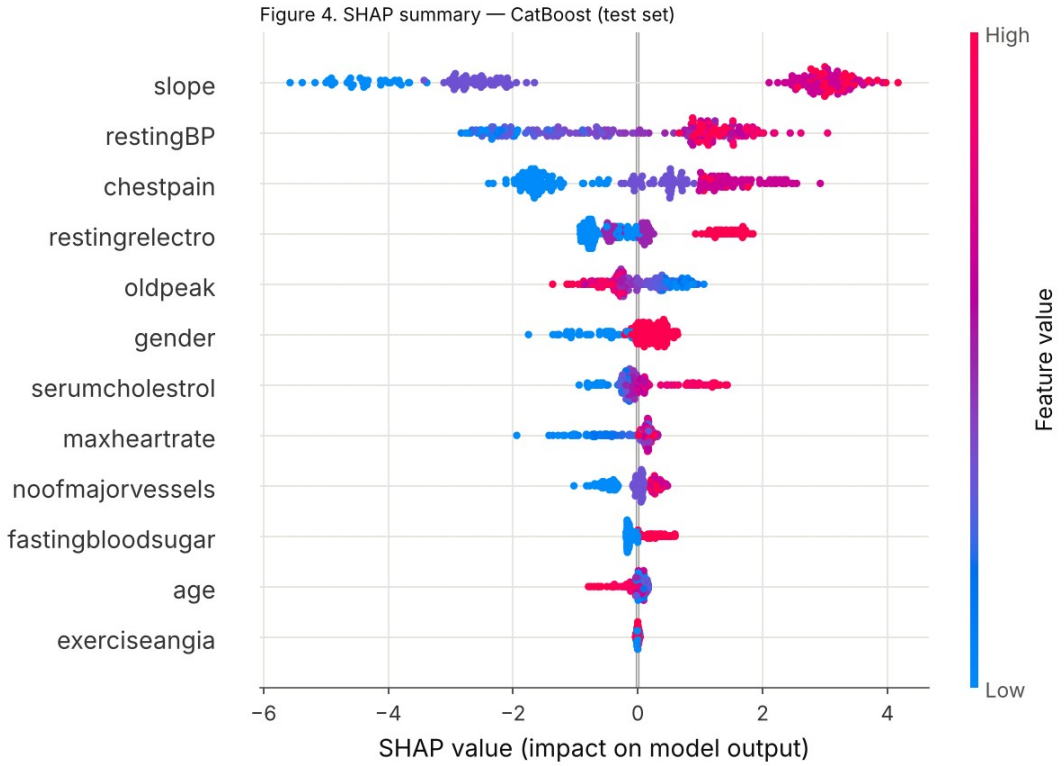


Figure 4. SHAP summary for CatBoost on the test set

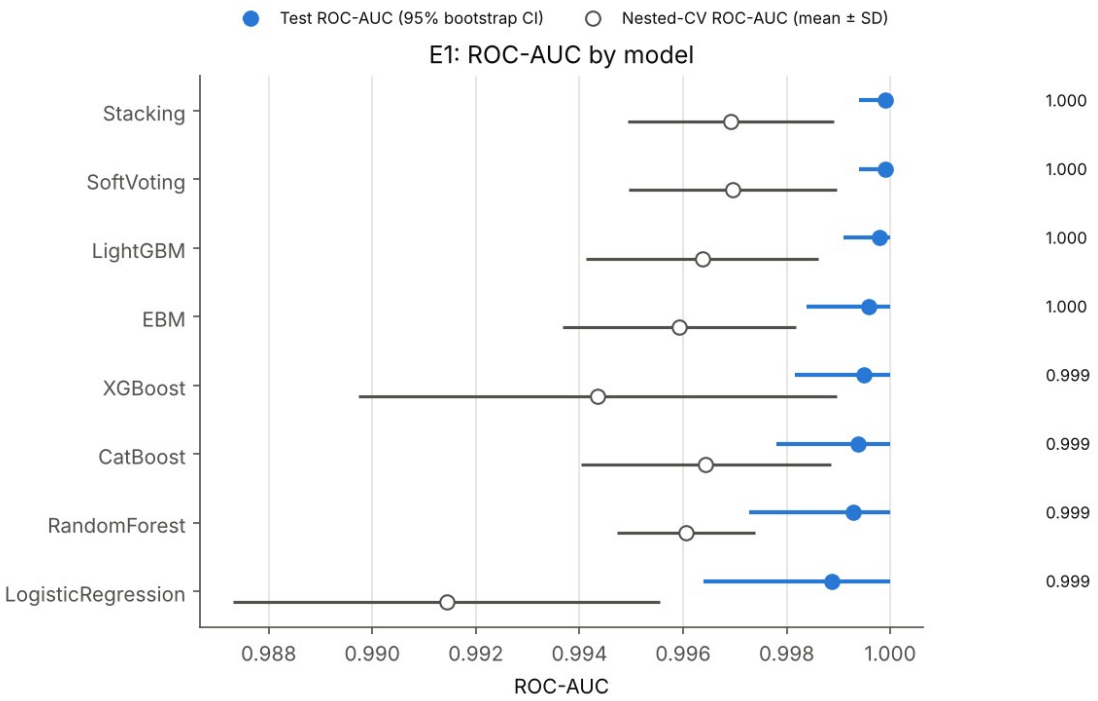


Figure 5. ROC-AUC by model: test set (95% bootstrap CI) and nested cross-validation (mean ± SD)

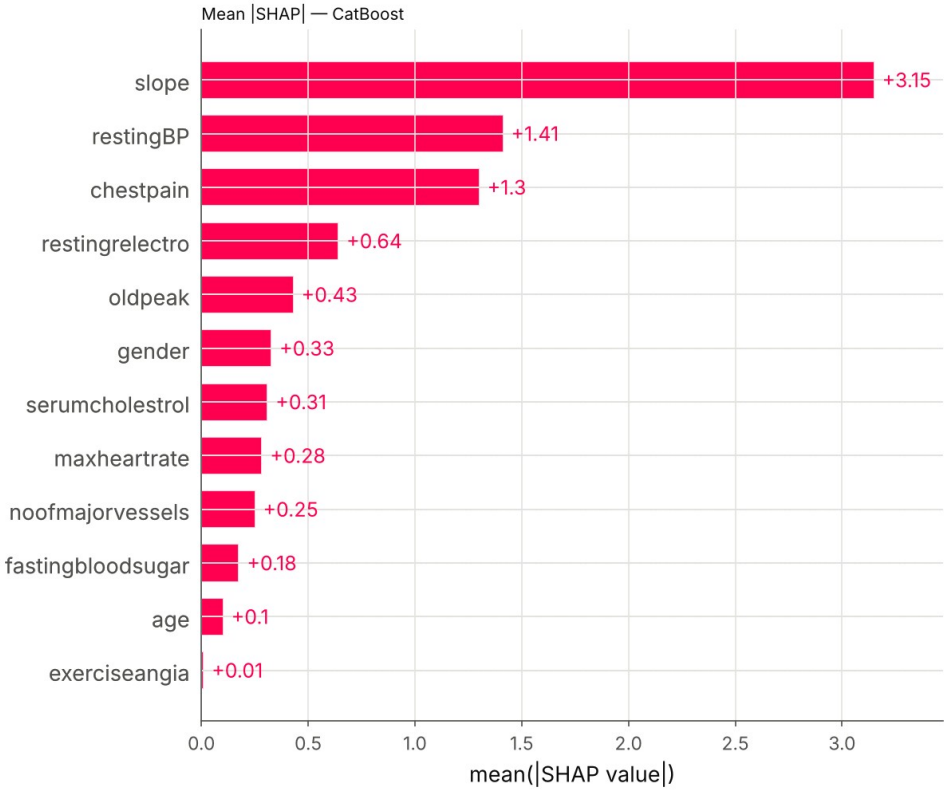


Figure 6. Mean absolute SHAP value per attribute (CatBoost)

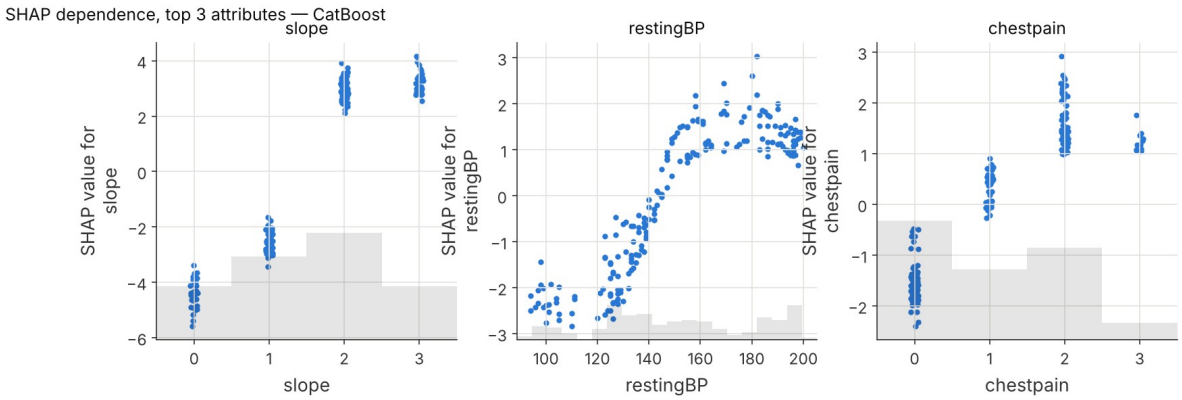


Figure 7. SHAP dependence plots for the three most important attributes (CatBoost)



Figure 8. EBM shape functions for the three most important attributes

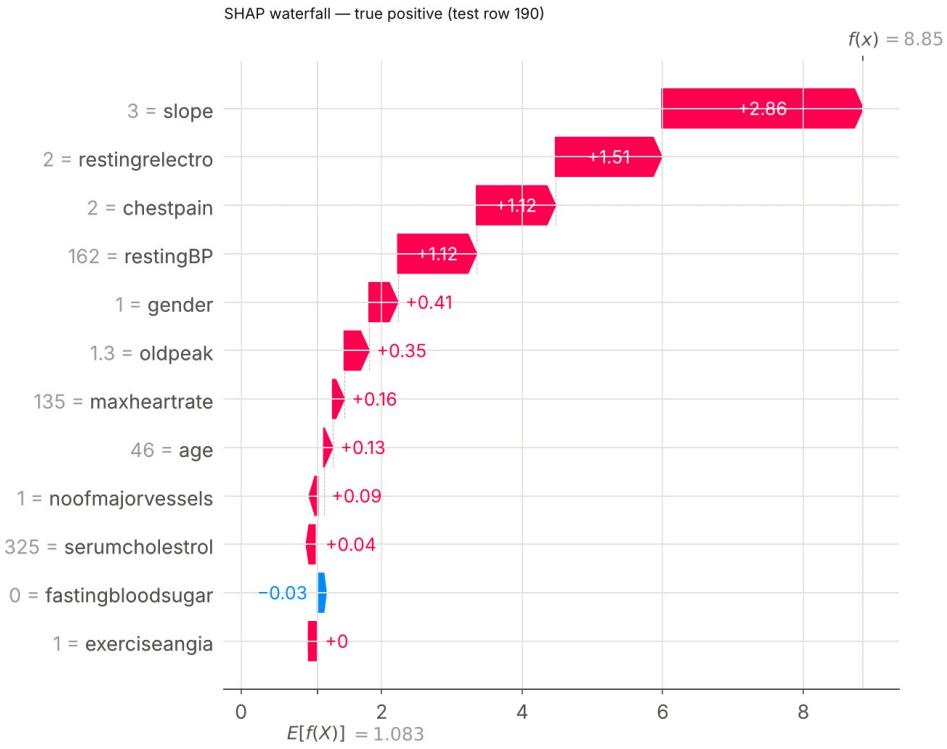


Figure 9. SHAP explanation for a correctly detected patient (true positive)

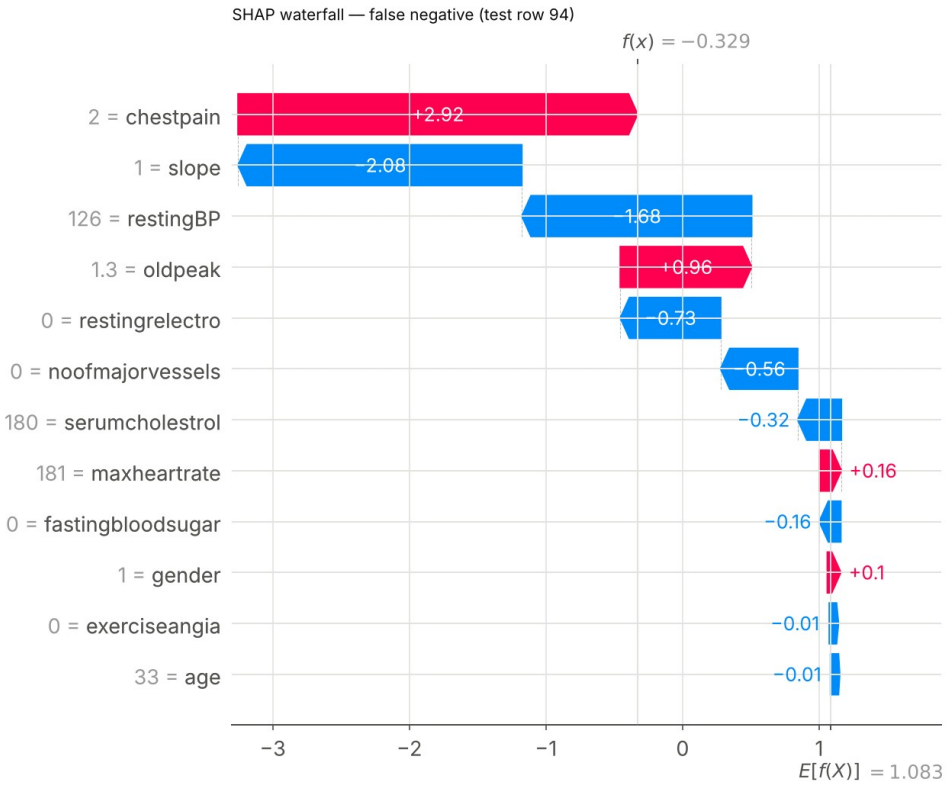


Figure 10. SHAP explanation for a missed patient (false negative)